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PROJECT DESCRIPTION

- **Background**

This analysis project focuses on the educational technology company, Educative, known for its online platform that offers a diverse range of courses to empower individuals seeking to enhance their skills in both professional and academic domains.

The dataset spans a comprehensive 7-year period, including data points from **2011 (July, September, November-December), the entire year of 2012-2016, and the first half of 2017 (January - July)**. Within this dataset, four distinct subjects emerge: **Web Development (W.D.), Graphic Design (G.D.), Musical Instruments (M.I.), and Business Financing (B.F.)**. Each subject undergoes further categorization into four proficiency levels: All, Beginner, Intermediate, and Expert.

- **Business Problem**

The Head of Curriculum at Educative has tasked the data analyst with presenting an overview of course revenue, aiming to identify opportunities for revenue growth and assess the performance of various courses. The team has been provided with comprehensive data across different topics to facilitate this analysis.

Notably, the manager has expressed a particular interest in Web Development, given its widespread popularity. The manager strongly advocates for an increase in course fees within the Web Development category to capitalize on its demand and enhance revenue streams for Educative.

DATA DESIGN

The data for this analysis project is structured across four CSV files, each representing one of the four subjects (Web Development, Graphic Design, Musical Instruments, Business Financing). These files collectively encompass a 7-year timeframe. The data collection period per year is as follows:

1. 2011: **July, September, November-December**
2. 2012-2016: **Full year**
3. 2017: **January - July**

This chronological breakdown ensures a comprehensive coverage of information for the specified subjects over the specified duration, facilitating a thorough analysis of trends and patterns.

All the data given per subject had the following columns

- **course_id** (Had unique identifiers for each course record)
 - **course_title** (These are the names of each course)
 - **url** (The website link to the courses)
 - **price** (The price per course)
 - **num_subscribers** (The number of individuals who had subscribed to the course)
 - **num_reviews** (The number of reviews given per course)
 - **num_lectures** (The number of lectures per the course)
 - **level** (Level of each course which is either Beginner/Intermediate/Expert/All)
 - **Rating** (Rating for each course which span from 0-1)
 - **content_duration** (The length of each course measured in hours)
 - **published_timestamp** (The date and time with which the courses were published)
 - **subject** (Categorization of courses to either Web Development/Graphic Design/Musical Instruments/Business Financing)
-
- **Data Cleaning**

In the data cleaning process, the following steps were undertaken:

1. Merging CSV Files:

- Combined the information from the four CSV files into a single file, resulting in a unified dataset with one sheet.

- The merged dataset now comprises 3682 records, consolidating information across the various subjects.

2. Duplication of Merged Data:

- A duplicate of the original merged data was created and stored in another sheet.
- This duplicate serves as a backup or reference point, ensuring the preservation of the initial dataset.

3. Deletion of Default Column:

- The "Source.Name" column, which likely contained default or redundant information, was removed from the dataset.
- This action streamlines the dataset, eliminating unnecessary columns and potentially improving the clarity and efficiency of subsequent analyses.

Source Name	Course ID	Course Title
Data_Sheet_Education_Courses..._Business_Courses.xlsx	48798	Bitcoin or Hype? Learn to Stop
Data_Sheet_Education_Courses..._Business_Courses.xlsx	48801	Accounting in 60 Minutes - A Big
Data_Sheet_Education_Courses..._Business_Courses.xlsx	133336	Stock Market Investing for Dummies
Data_Sheet_Education_Courses..._Business_Courses.xlsx	133336	Introduction to Financial Markets
Data_Sheet_Education_Courses..._Business_Courses.xlsx	648839	The Complete Financial Analyst
Data_Sheet_Education_Courses..._Business_Courses.xlsx	93338	Forex Basics
Data_Sheet_Education_Courses..._Business_Courses.xlsx	321439	Beginner to Pro in Swift Forex
Data_Sheet_Education_Courses..._Business_Courses.xlsx	321442	Stocks and Trading: Build Your 10
Data_Sheet_Education_Courses..._Business_Courses.xlsx	189100	Financial Analysis: A Recipe for S
Data_Sheet_Education_Courses..._Business_Courses.xlsx	133336	Stock Market Fundamentals
Data_Sheet_Education_Courses..._Business_Courses.xlsx	754028	Financial Accounting - A Brief Int
Data_Sheet_Education_Courses..._Business_Courses.xlsx	202560	Fundamentals of Forex Trading
Data_Sheet_Education_Courses..._Business_Courses.xlsx	208080	Forex Trading A-Z: 1001 Tradi
Data_Sheet_Education_Courses..._Business_Courses.xlsx	285438	Learn to Trade for Profit: Trading
Data_Sheet_Education_Courses..._Business_Courses.xlsx	402384	Options Trading Introduction: Th
Data_Sheet_Education_Courses..._Business_Courses.xlsx	189996	Introduction to Digital Payments
Data_Sheet_Education_Courses..._Business_Courses.xlsx	302382	Introduction to Accounting: The
Data_Sheet_Education_Courses..._Business_Courses.xlsx	777884	Corporate Finance - A Brief Intro
Data_Sheet_Education_Courses..._Business_Courses.xlsx	133336	CFO 101: How To Master Offici
Data_Sheet_Education_Courses..._Business_Courses.xlsx	488716	Introduction to Finance, Investm
Data_Sheet_Education_Courses..._Business_Courses.xlsx	778374	Options for Beginners: Your Step
Data_Sheet_Education_Courses..._Business_Courses.xlsx	321834	Learn to Trade for Profit: Find a
Data_Sheet_Education_Courses..._Business_Courses.xlsx	208080	Forex Basics: Master the Game 10
Data_Sheet_Education_Courses..._Business_Courses.xlsx	208080	Beginner Binary Options Course
Data_Sheet_Education_Courses..._Business_Courses.xlsx	108802	Finance: Rapid Analysis & Invest
Data_Sheet_Education_Courses..._Business_Courses.xlsx	508308	Create A Business From Home 1
Data_Sheet_Education_Courses..._Business_Courses.xlsx	413338	Options Trading Basics 10 Course
Data_Sheet_Education_Courses..._Business_Courses.xlsx	645100	Accounting & Financial Statement
Data_Sheet_Education_Courses..._Business_Courses.xlsx	133336	Improve Your Financial Literacy
Data_Sheet_Education_Courses..._Business_Courses.xlsx	705862	Forex Trading For Beginners
Data_Sheet_Education_Courses..._Business_Courses.xlsx	103864	Stock Trading Ninja: Complete S

4. Handling Blank Rows:

- Cleaned the dataset by removing blank rows based on the "course_id" column, ensuring data integrity.

5. Column Header Contextualization:

- Renamed all column headers to provide clearer and more descriptive context for each attribute, enhancing the dataset's interpretability.

6. Duplicate Rows Removal:

- Identified and removed duplicate rows, resulting in a streamlined dataset with 3676 records.

7. Timestamp Manipulation:

- Created two new columns from the "published_timestamp" column by separating the date and time using a delimiter (space). The original timestamp data was then deleted.

8. Published Period Categorization:

- Utilized the newly created "published time" column to categorize the publishing period into Morning, Afternoon, or Evening.

9. Subject Standardization:

- Replaced instances of "Subject: Web Development" in the subject column with the standardized term "Web Development."

10. Free or Paid Indication:

- Introduced a new column, "Free or Paid," which utilized the "price" column to denote whether a course is free (0\$) or paid.

11. Revenue Calculation:

- Created a "Revenue" column, calculating the revenue generated by each course using the formula: $\text{Revenue} = \text{Price} * \text{Number of Subscribers}$.

12. Data Type Adjustments:

- Modified the data types of specific columns: Course ID changed from Number to Text, Price and Revenue from Number to Currency, and Published Time from Datetime to Time.

13. Top 20 Most Subscribed Courses:

- Generated a new sheet highlighting the top 20 most subscribed courses using conditional formatting. This information was complemented with additional details such as Level, Free/Paid status, Content Duration, and Published Date obtained through a VLOOKUP.

14. Pivot Tables and Visualizations:

- Created pivot tables to summarize data for visualization purposes, employing various chart types:
 - a. Total number of subscribers for each subject (Pie Chart)
 - b. Average subscriber count per subject (Bar Chart)
 - c. Average rating per subject for each level (Column Chart)
 - d. Average cost per subject for each level (Column Chart)
 - e. Average Content Duration per subject (Bar Chart)
 - f. Annual Revenue (Line Graph with Markers)

VISUALIZATIONS

The selection of visualization tools is crucial for effectively conveying insights from the data analysis. In this project, two tools were chosen:

a. Excel:

Excel was chosen due to its widespread familiarity and ease of use among stakeholders. Many individuals are already familiar with Excel, making it accessible for a broader audience. Its versatility in creating various charts and tables also contributes to its effectiveness for data analysis.

b. Power BI:

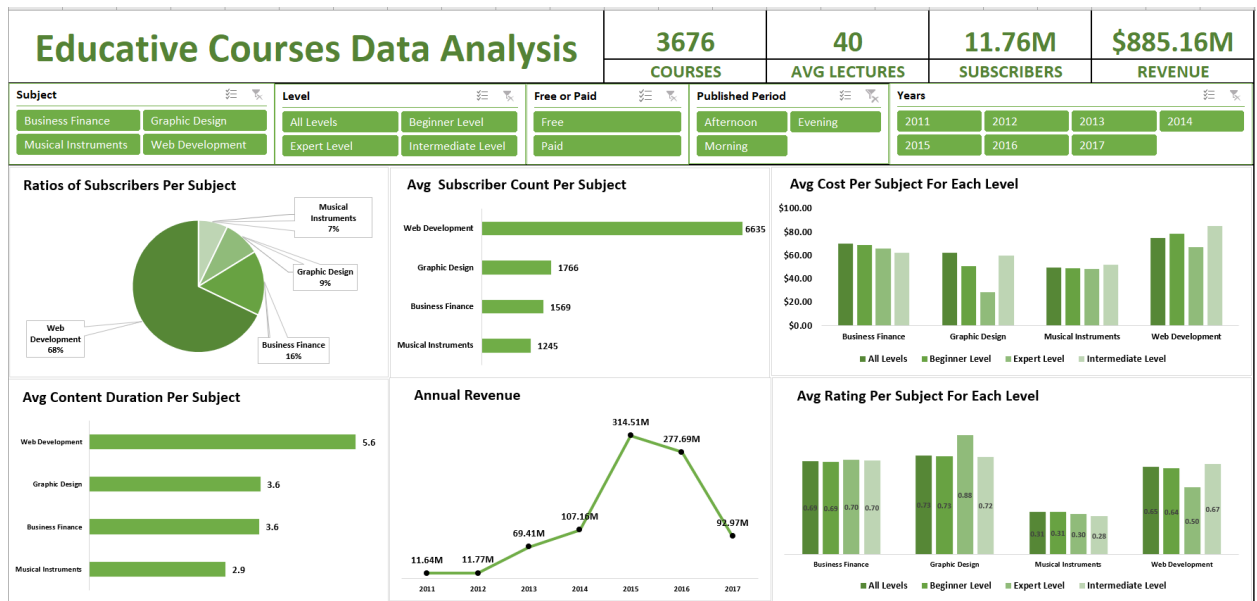
Power BI was selected for its ability to create interactive dashboards and scalability. This tool offers dynamic visualization capabilities, allowing users to interact with the data and explore different perspectives. Its scalability makes it suitable for handling large datasets and providing a more comprehensive view of the information.

Additional Features:

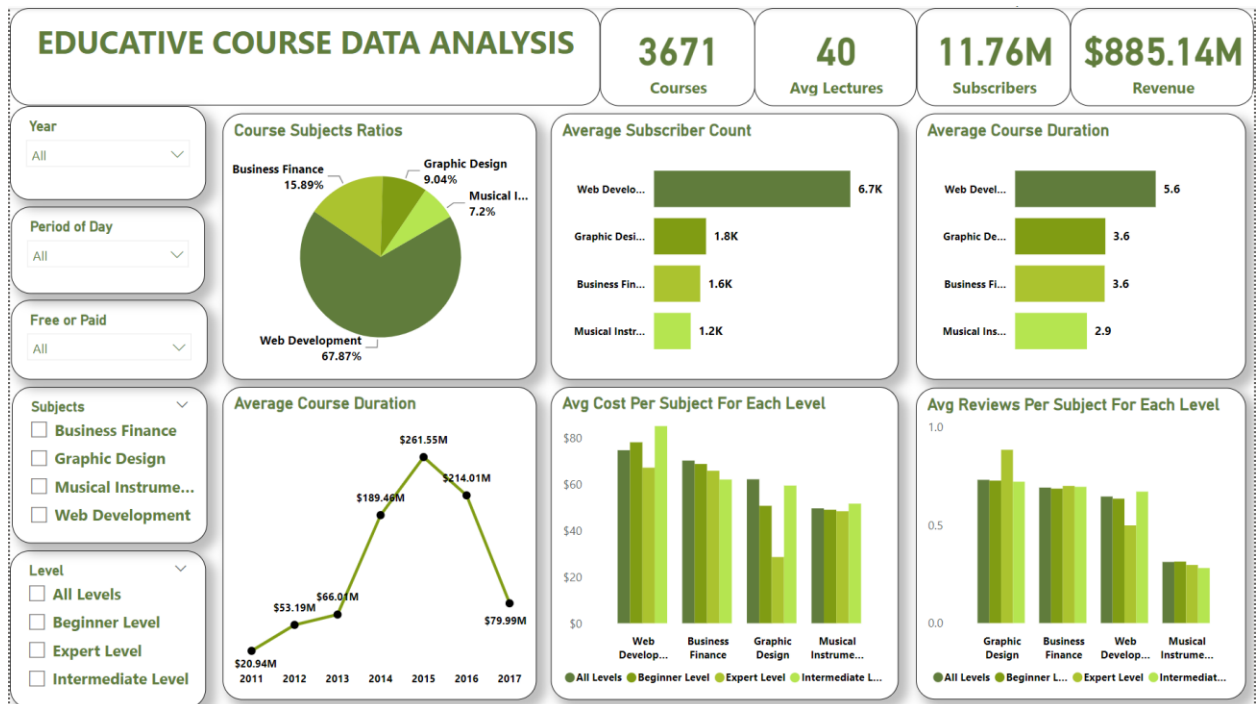
- **Slicers:** Implemented slicers to simplify data filtering and exploration, enhancing user experience and making it more intuitive to analyze specific subsets of the data.
- **Cards:** Utilized cards to display key metrics or summarized information in a concise and easily digestible format. Cards are effective in highlighting important figures or trends, providing a quick overview of essential insights.

By combining the strengths of Excel and Power BI, along with the interactive features like slicers and cards, the visualization strategy ensures that stakeholders can efficiently explore and understand the findings derived from the data analysis.

Visualization from Excel



Visualization from Power BI



FINDINGS

In the below account, the following words will be referred to as their aliases which are in the brackets - *Business Finance (B.F.)*, *Graphic Design (G.D.)*, *Music Instruments (M.I.)* and *Web Development (W.D.)*

Here are the findings of from the data given:

MARKET SHARE (USING TOTAL SUBSCRIBERS)

1. Web Development (W.D.)

- Dominated the course market consistently, holding an impressive **67.87%** market share across all years.
- Established itself as the leading subject, showcasing sustained popularity and demand.

2. Business Finance (B.F.)

Secured a substantial market share of **15.89%**, positioning itself as the second-largest course provider in 4 out of the 6 years (2013-2016).

- Demonstrated a consistent presence and relevance in the course market.

3. Graphic Design (G.D.)

- Held a market share of **9.04%**, experiencing its highest position as the third-largest course provider in the years 2012, 2014-2016.
- Maintained a steady presence, contributing to the diversity of the course offerings.

4. Musical Instruments (M.I.)

- Accounted for a market share of **7.2%**, consistently trailing in course market share for 4 consecutive years (2014-2017).
- While having a smaller market share, M.I. continued to contribute to the overall course portfolio.

Overall Implications

- W.D. dominates the market, signifying its sustained popularity.
- B.F. maintains a strong position as the second-largest course provider.
- G.D. and M.I. contribute to course diversity, each holding a significant share of the market.

AVERAGE SUBSCRIBER ENGAGEMENT

1. Top Subjects in Average Subscribers (2012-2016):

- Web Development (W.D.) consistently stood out as the top subject in terms of average subscribers over the 7-year period. Graphic Design (G.D.) closely followed as the second-highest, maintaining this position for four consecutive years (2014-2017).

2. Average Subscribers by Proficiency Levels:

All Levels:

- Web Development (W.D.) consistently secured the top position in average subscribers across all proficiency levels.
- Graphic Design (G.D.) consistently ranked second in average subscribers across all proficiency levels

Beginner Level:

- Web Development (W.D.) and Graphic Design (G.D.) claimed the first and second positions, respectively, in average subscribers for beginner-level courses.

Intermediate and Expert Levels

- Business Finance (B.F.) and Musical Instruments (M.I.) took the second and third positions in average subscribers for both intermediate and expert levels.

3. Observations on Subject Performance

Web Development (W.D.):

- Dominated across all proficiency levels, securing the highest average subscribers.
- Maintained a consistent lead over other subjects, showcasing sustained popularity.

Graphic Design (G.D.):

- Established itself as a strong performer, consistently ranking second in average subscribers across all levels
- Demonstrated a notable and sustained appeal to subscribers.

Business Finance (B.F.) and Musical Instruments (M.I.)

- Positioned as the second and third choices for intermediate and expert levels, showcasing specific areas of interest within these subjects.

AVERAGE CONTENT DURATION

Average Content Durations (Ascending):

- Musical Instruments (M.I.): 2.8 hours
- Business Finance (B.F.) and Graphic Design (G.D.): 3.6 hours
- Web Development (W.D.): 5.6 hours

Educative can use this duration breakdown to strategically plan and market courses, catering to diverse learner preferences and schedules.

These insights not only provide numerical values but also offer practical considerations for learners and course planners, enhancing the overall usability of the duration analysis.

AVERAGE COST PER SUBJECT FOR EACH LEVEL

1. Costliest Courses by Subject and Level

- Business Finance (B.F.) and Graphic Design (G.D.) exhibit their highest average course costs in the All Level category.
- Musical Instruments (M.I.) and Web Development (W.D.) feature their most costly courses at the Intermediate Level.

2. Expert Level Cost Trends

Across all four subjects, Expert level courses are consistently the least costly on average, maintaining affordability for learners.

3. Average Costs by Level

- | | |
|----------------------------------|--------------------------------------|
| • All Level: \$66.75 | • Intermediate Level: \$58.02 |
| • Beginner Level: \$65.24 | • Expert Level: \$66.94 |

4. Overall Average Cost

The overall average cost for all proficiency levels is \$66.11.

AVERAGE RATING PER SUBJECT FOR EACH LEVEL

Across all subjects, levels, and years, the average rating stands impressively at 0.61 on a scale of 0-1, indicating positive overall feedback.

1. Paid vs. Free Courses

On average, paid courses received a slightly higher rating of 0.60, compared to free courses with a rating of 0.59.

2. Rating Averages by Proficiency Level

- All Level: 0.62
- Beginner Level: 0.60
- Intermediate Level: 0.59
- Expert Level: 0.6

3. Subject-Specific Ratings

- G.D.: 0.73
- B.F.: 0.69
- W.D.: 0.64
- M.I.: 0.31

4. Subject and Level Breakdown:

- Business Finance (B.F.): Highest average rating (0.70) in Intermediate and Expert levels.
- Graphic Design (G.D.): Expert Levels achieved the highest rating of 0.88.
- Musical Instruments (M.I.): All levels and Beginner levels had the highest average rating of 0.31.
- Web Development (W.D.): Intermediate Level garnered the highest average rating of 0.67.

REVENUE

Over the provided period, a total of \$885.16M in revenue was generated.

Revenue demonstrated consistent growth from 2011 to 2015, followed by a substantial drop in 2016. This decline is attributed to presumed market saturation, and subsequent years show a yet-to-recover trend.

1. Subject Market Share Breakdown

- Web Development (W.D.) dominated the market with a substantial 71.30% share, followed by Business Finance (B.F.) at 13.98%, Graphic Design (G.D.) at 8.70%, and Musical Instruments (M.I.) at 6.03%.

2. Revenue Upsurge Periods for Each Subject

- B.F.: Experienced a revenue upsurge from 2014 to 2016
- G.D.: Witnessed revenue growth in 2015-2016.
- M.I.: Had a revenue spike in 2014.
- W.D.: Experienced revenue growth in 2015-2016.

3. Level-wise Revenue Generation:

- Across all subjects, All Levels and Beginner Level consistently generated the highest revenue.

4. Subscriber Impact on Revenue

- Positive correlation observed between subscriber data and revenue generation, emphasizing the influence of subscriber engagement on overall revenue.

CONCLUSION

The analysis undertaken to address the business question posed by the Head of Curriculum at Educative has provided valuable insights into course revenue and performance across various subjects. The focus on Web Development (W.D.) aligns with its consistent dominance in the course market, holding an impressive 71.30% share. This popularity underscores its sustained demand, validating the manager's interest in increasing course fees within this subject.

Key Findings

Market Share Dominance

- Web Development (W.D.) has consistently dominated the course market, securing a substantial 67.87% to 71.30% market share over the 7-year period.
- Business Finance (B.F.) maintained a strong second position, contributing significantly to Educative's course offerings.

2. Average Subscriber Engagement

- W.D. consistently attracted the highest average subscribers across all proficiency levels, emphasizing its sustained appeal.
- Graphic Design (G.D.) positioned as a strong performer, consistently ranking second in average subscribers.

3. Content Duration and Cost Analysis

- Courses in Web Development (W.D.) generally require a more substantial time commitment, with an average content duration of 5.6 hours.
- Business Finance (B.F.) and Graphic Design (G.D.) feature their costliest courses at the All Level category, indicating a balance between depth and accessibility.

4. Rating Insights

- Courses across all subjects received an impressive average rating of 0.61 on a scale of 0-1
- Positive correlation observed between subscriber data and revenue generation, emphasizing the importance of subscriber engagement.

5. Revenue Trend

- Over the provided period, Educative generated \$885.16M in revenue, showcasing consistent growth until 2015, followed by a decline in 2016
- Web Development (W.D.) experienced revenue growth in 2015-2016, aligning with its continued market dominance.

Recommendations

Web Development Emphasis: Considering W.D.'s sustained popularity, the recommendation aligns with the manager's proposal to increase course fees within this subject, capitalizing on its strong market position.

Diversification Strategies: While W.D. remains a stronghold, exploring strategies to enhance offerings in Business Finance (B.F.) and Graphic Design (G.D.) could further diversify revenue streams and cater to a broader audience.

Continuous Monitoring: Regular monitoring of market trends, subscriber engagement, and competitor activities will be crucial for Educative to adapt swiftly and stay ahead in the dynamic online education landscape.

In conclusion, the analysis not only provides a comprehensive understanding of Educative's course landscape but also offers actionable insights to inform strategic decisions, drive revenue growth, and ensure the continued success of the online education platform.